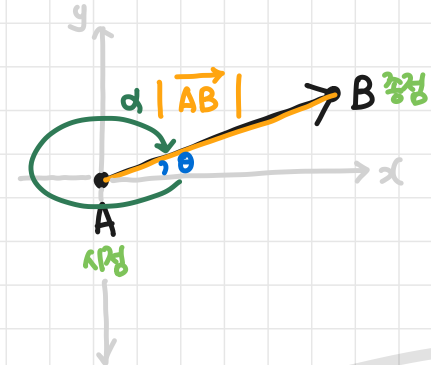
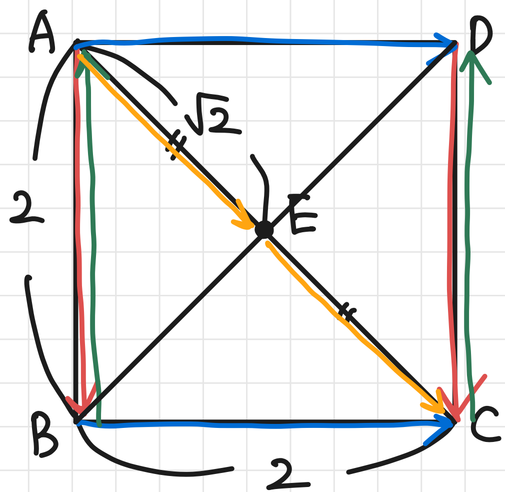
• 벡터

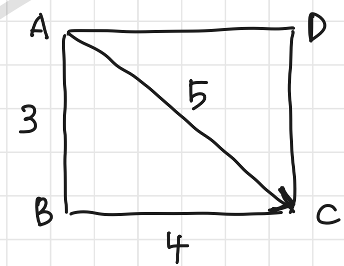
- ○ 스칼라 vs 벡터 : 크기 + 방향. → 물리학.
- ○ 두 점을 지나는 화살표다.
- ○ 벡터의 크기 : $|\vec{AB}|$
- ○ 벡터의 방향 : θ



$\vec{AB} = \vec{a}$



#예16-1.



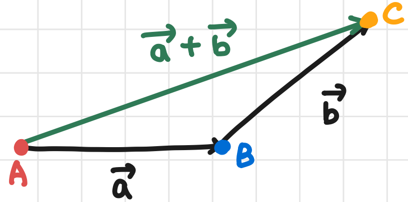
$\vec{AC} = 5.$

#예16-2.

벡터는 점이다. e.g.) $\vec{a} = (2,3)$
$\vec{b} = (7,1)$

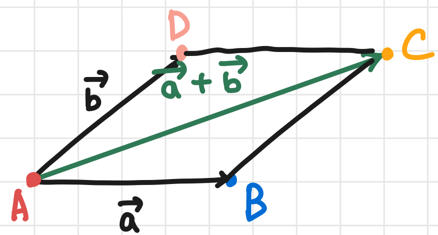
- 덧셈, 뺄셈, 실수배(상수배), 내적, 외적

- 벡터의 덧셈

○ 

$\vec{a} + \vec{b} =$

$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$

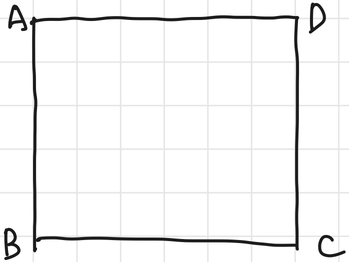
○ 

$\overrightarrow{AB} + \overrightarrow{AD} = \overrightarrow{AC}$.

○ $\vec{a} + \vec{b} = \vec{b} + \vec{a}$

○ $(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$

• 벡터의 덧셈에 대한 항등원 : 영벡터 ($\vec{0}$)



$\vec{AA} = \vec{BB} = \vec{CC} = \vec{DD}$

$= \vec{0}$

예) $\vec{AB} + \vec{AA} = \vec{AB}$

$\vec{AB} + \vec{0} = \vec{AB}$

$\vec{a} + \boxed{} = \vec{a}$

$(2,3) + \boxed{} = (2,3)$

$(0,0) = \vec{0}$ = 영벡터

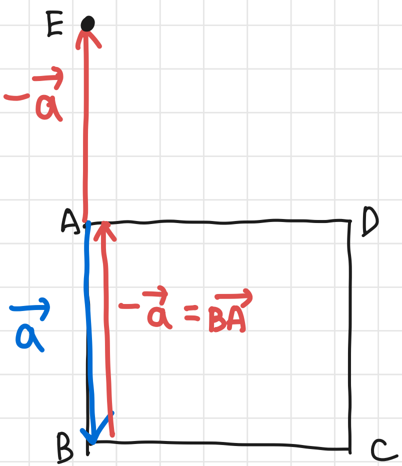
Recap) 항등함수 $y = x$. (I).

$y = I(x)$

O으로부터 A=(x,y) 길이

: $\sqrt{x^2+y^2}$

• 벡터의 덧셈에 대한 역원 : $(-\vec{a})$



$\vec{a} = \vec{AB}$

$-\vec{a} = -\vec{AB} = \vec{BA}$

$\vec{a} + \boxed{} = \vec{0}$

$(2,3) + (\underset{-2,-3}{\quad\quad}) = (0,0)$

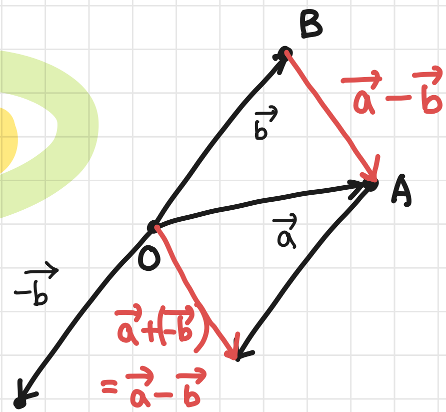
$-(2,3) = -\vec{a}$

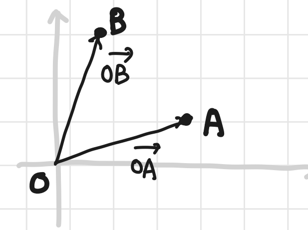
$|-\vec{a}| = |\vec{a}| = \sqrt{2^2+3^2} = \sqrt{4+9} = \sqrt{13}$

- 벡터의 뺄셈

$\vec{a} = \overrightarrow{OA}$, $\vec{b} = \overrightarrow{OB}$ 일 때, $\vec{a} - \vec{b} = ?$

$$\overrightarrow{OA} - \overrightarrow{OB} = \overrightarrow{BA}$$



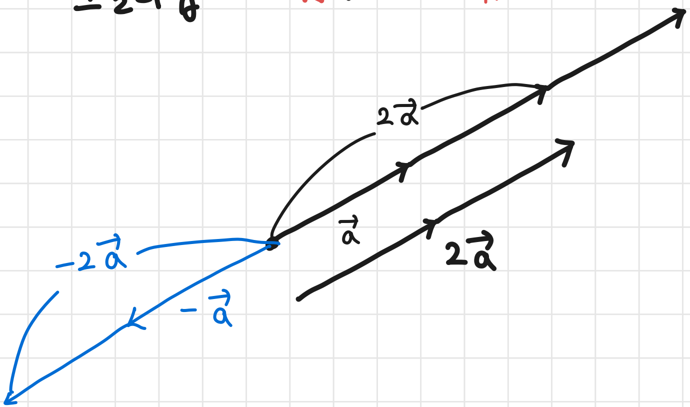


$$\overrightarrow{CA} - \overrightarrow{CB} = \overrightarrow{BA}$$

$$\overrightarrow{CA} - \overrightarrow{DB} = \overrightarrow{CA} - (\overrightarrow{DC} + \overrightarrow{CB})$$

$$= \overrightarrow{CA} - \overrightarrow{CB} - \overrightarrow{DC} = \overrightarrow{BA} - \overrightarrow{DC}$$

- 스칼라곱: $k\vec{a}$ (k는 실수)



① 실수배

② $\vec{b} = k\vec{a}$ 이면 $\vec{a}, \vec{b}$는 평행하다.

① $k>0$ 이면, $\vec{a}$랑 방향 같고, $|\vec{a}|$보다 k배인 크기.

② $k<0$ 이면, 〃 방향 반대고, $|\vec{a}|$보다 k배인 크기